

Hand on Unit-3 Descriptive Statics

Program 1:

Output:

Number of Rows : 10

Number of Columns : 6

Data Types

EmployeeID int64

Name object

Age float64

Department object

Salary float64

Experience float64

dtype: object

Missing Values

EmployeeID 0

Name 0

Age 1

Department 0

Salary 1

Experience 1

dtype: int64

Statistical Summary

	EmployeeID	Name	Age	Department	Salary	Experience
count	10.000000	10	9.000000	10	9.000000	9.000000
unique	NaN	10	NaN	4	NaN	NaN
top	NaN	Alice	NaN	HR	NaN	NaN
freq	NaN	1	NaN	3	NaN	NaN

mean	105.50000	NaN	29.222222	NaN	46444.444444	5.111111
std	3.02765	NaN	3.153481	NaN	7401.201104	1.900292
min	101.00000	NaN	25.000000	NaN	35000.000000	2.000000
25%	103.25000	NaN	27.000000	NaN	42000.000000	4.000000
50%	105.50000	NaN	29.000000	NaN	47000.000000	5.000000
75%	107.75000	NaN	31.000000	NaN	50000.000000	6.000000
max	110.00000	NaN	35.000000	NaN	60000.000000	8.000000

Program 2:

Output:

Grade

A 7

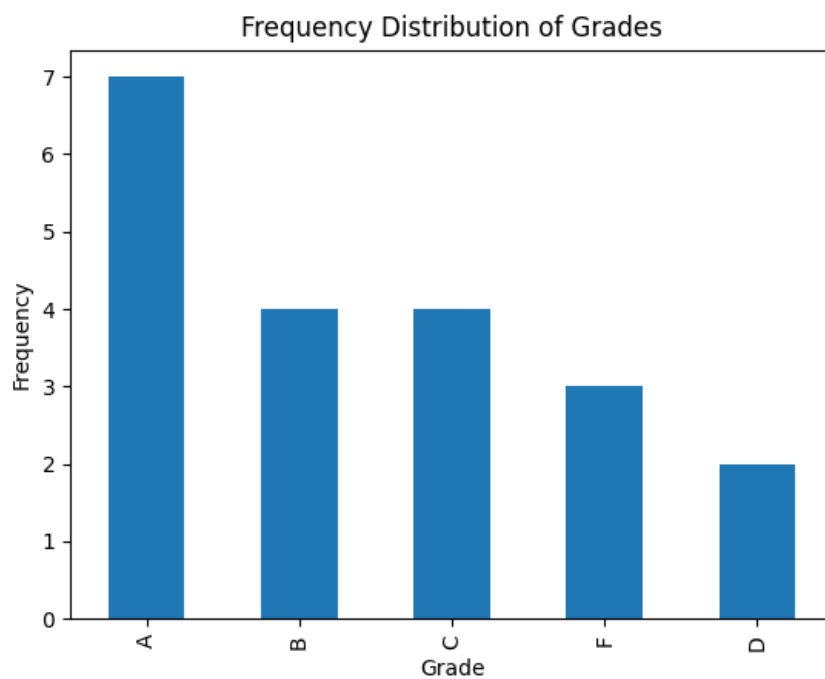
B 4

C 4

F 3

D 2

Name: count, dtype: int64



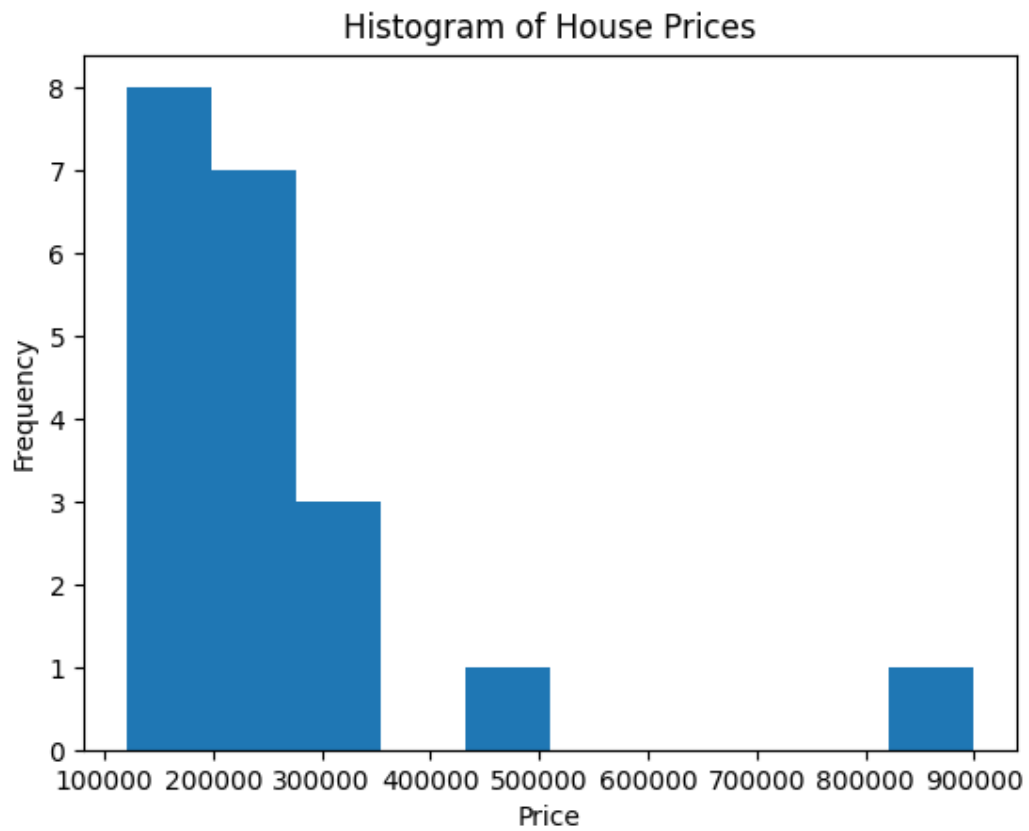
Program 3:

Output:

Mean : 262000.0

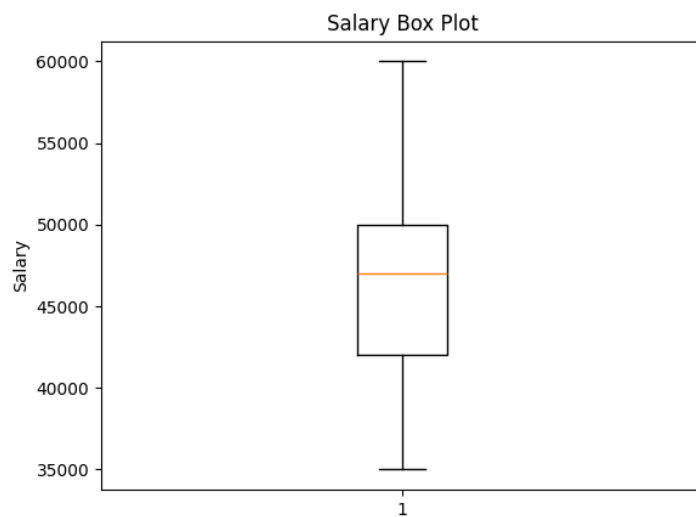
Median : 225000.0

Right Skewed Distribution



Program 4:

Output:



Program 5:

Output:

Outliers

	StudentID	Name	Marks	Grade
12	13	Mary	35	F
18	19	Sam	20	F
19	20	Tom	150	A

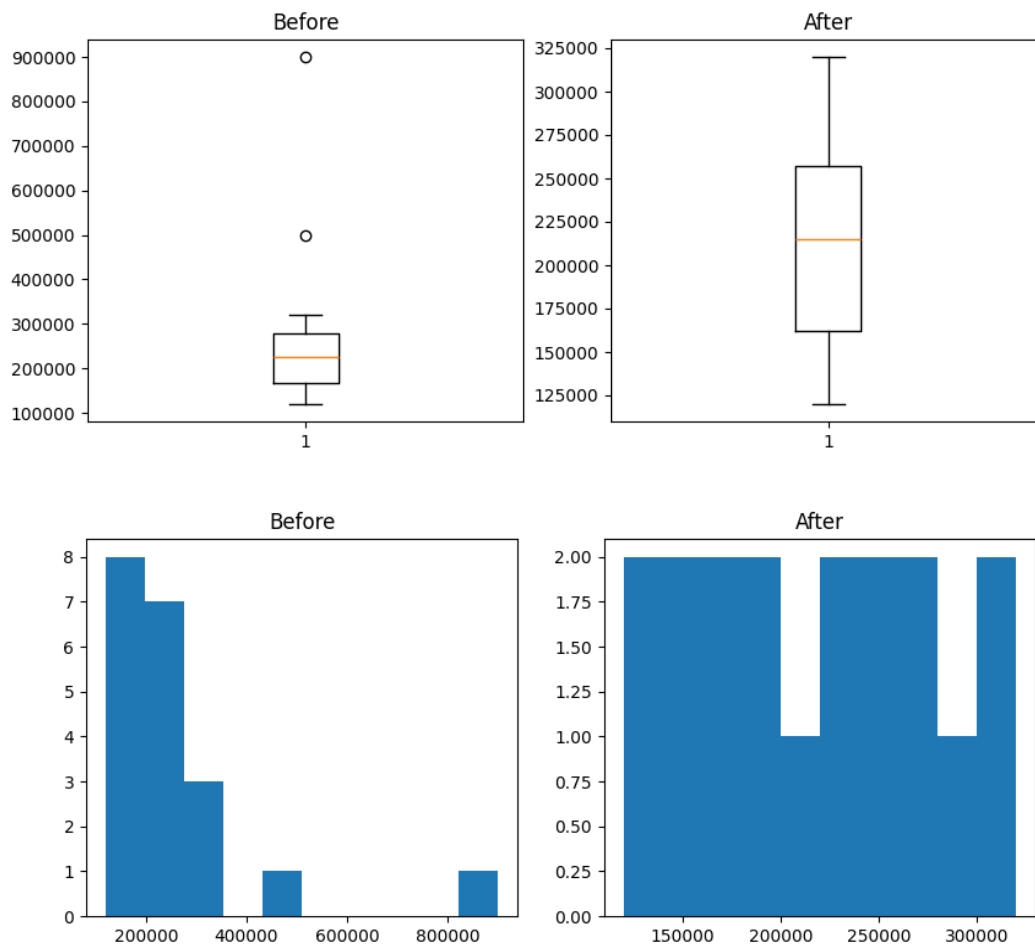
Program 6:

Output:

	HouseID	Price
0	1	120000
1	2	135000
2	3	140000
3	4	150000
4	5	160000
5	6	170000
6	7	180000
7	8	190000
8	9	210000
9	10	220000
10	11	230000
11	12	240000
12	13	250000
13	14	260000
14	15	275000
15	16	290000
16	17	300000
17	18	320000

Program 7:

Output:



Program 8:

Output:

RangeIndex: 891 entries, 0 to 890

Data columns (total 12 columns):

#	Column	Non-Null Count	Dtype
0	PassengerId	891 non-null	int64
1	Survived	891 non-null	int64
2	Pclass	891 non-null	int64
3	Name	891 non-null	object
4	Sex	891 non-null	object
5	Age	714 non-null	float64

```
6 SibSp      891 non-null  int64
7 Parch      891 non-null  int64
8 Ticket     891 non-null  object
9 Fare       891 non-null  float64
10 Cabin     204 non-null  object
11 Embarked   889 non-null  object
```

dtypes: float64(2), int64(5), object(5)

memory usage: 83.7+ KB

None

===== Missing Values =====

```
PassengerId    0
Survived        0
Pclass          0
Name            0
Sex             0
Age            177
SibSp           0
Parch           0
Ticket          0
Fare            0
Cabin          687
Embarked        2
dtype: int64
```

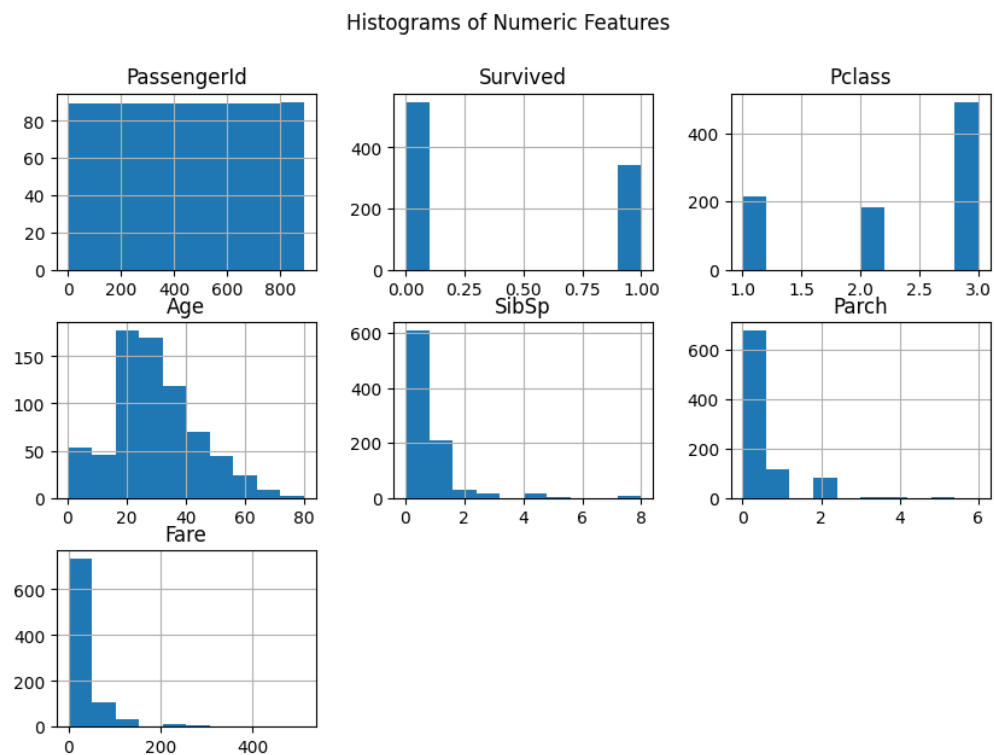
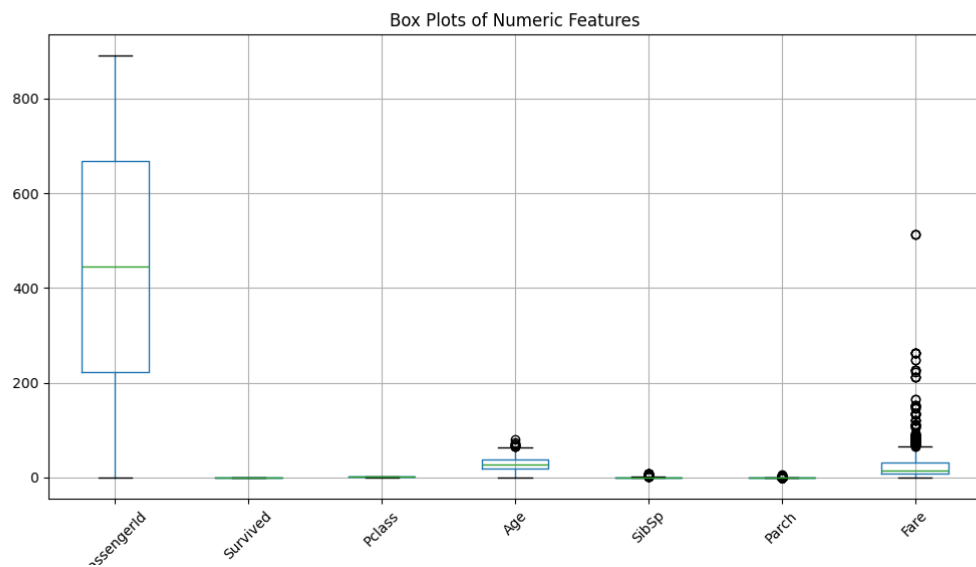
===== Descriptive Statistics =====

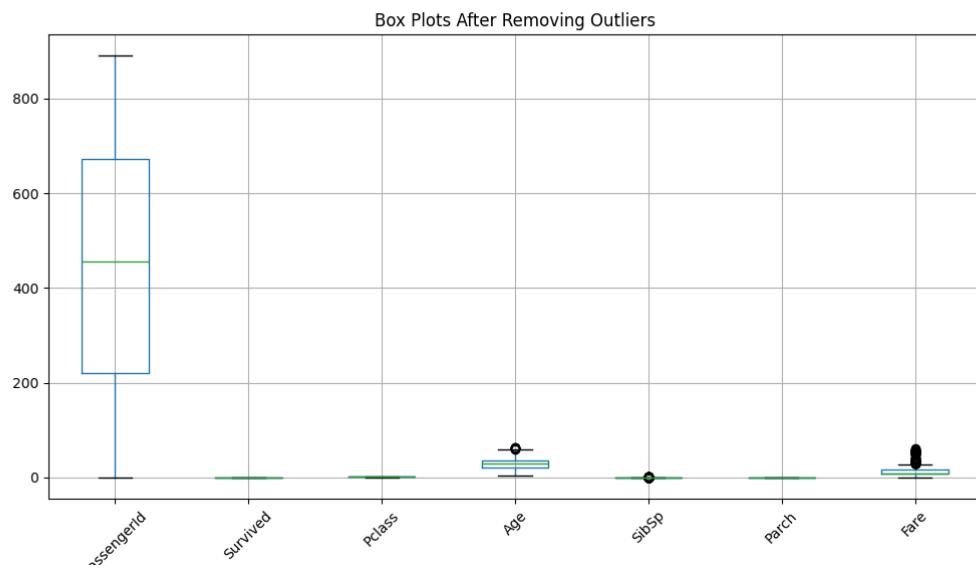
	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000

25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

Original Dataset Shape: (891, 12)

Cleaned Dataset Shape: (598, 12)





Histograms After Removing Outliers

